# Meiying Ren

Southern University of Science and Technology | Shenzhen, China
12432500@mail.sustech.edu.cn


## EDUCATION

***Southern University of Science and Technology | Shenzhen, China***

M.S. Candidate in Materials Science and Engineering | 2024-Present

- Advisor: Prof. Chuanfei Guo
- GPA: 3.67/4 (Top 10%) | Rank: 8/114
- Courses: Mechanical Behavior of Materials; Flexible Electronic Materials; Microscopy Analysis and Testing; Advanced Materials Characterization.

***Chang'an University | Xi'an, China***

Bachelor of Engineering in New Energy Materials and Devices | 2020-2024

- Advisor: Prof. Lei Gou
- GPA: 3.78/4 (Top 10%) | Rank: 3/40
- Courses: Fundamentals of Materials Science; Circuit and Analog Electronics; Solid-State Physics; Physical Chemistry; Semiconductor Physics and Devices.

## PUBLICATIONS

**Ren, M.#,** Huang, T.#, Dong, S.#, Fang, W., Fang, S., Guo, H., Zhang, N., Jiang, P., Huang, T., Yu, X., & Guo, C. Flexible iontronic pressure-sensing patches for remote tactile interaction. Manuscript submitted to *Science Advances*.

Zhang, Y., Li, J., Hao, L., Chen, Y., **Ren, M.,** Li, D., & Gou, L. Performance of jujube-derived carbon-modified $MnO_2$ cathode materials for zinc–manganese batteries. *Chemical Engineering*, **50**(3), 25–29 (2022).

## RESEARCH EXPERIENCE

### Flexible Electronics and Wearable Tactile-Interaction Systems | 2025–Present

- Addressed signal instability associated with skin viscoelasticity and deformation through coordinated material, interface, and structural design, developing a conformal 48-channel iontronic patch with reduced motion artifacts and baseline drift.
- Optimized the grid-spacer architecture, achieving a sensitivity of 4.1 pF $kPa^{-1}$ within 20 kPa and stable performance over 5,000 loading cycles.
- Established pressure-to-stimulation encoding and remote tactile interaction; collaborated on Altium Designer- and STM32CubeIDE-based circuit development, signal acquisition, system integration, and interference mitigation.

### Low-Drift and Low-Hysteresis Poly(ionic liquid) Elastomers | 2024–2025

- Addressed drift and hysteresis caused by chain slippage, viscoelastic relaxation, and ionic redistribution through molecular and crosslinked-network design of [VBIM][TFSI]-based elastomers.
- Screened six formulations across a 1–20 wt% crosslinker range, limiting signal drift to <1% over 10 min and reducing loading–unloading hysteresis.

**Carbon-Fiber-Reinforced Guzheng Plectra | 2023–2024**

- Developed carbon-fiber-reinforced plectra to address the trade-off between wear resistance, mechanical strength, and acoustic quality.
- Quantified the fundamental and harmonic components using FFT, achieving 86.3% spectral similarity to the reference plectrum and enabling reproducible batch fabrication through matched-mold design.

## PATENTS

- A Grid-Structured Spacer for an Iontronic Pressure Sensor. | Pending
- A Tactile Interaction System and Its Control Method. | Pending
- A Carbon Fiber Composite Guzheng Plectrum. | CN116645935A
- A Metal Mold for Manufacturing Carbon Fiber Composite Guzheng Plectra. | CN221641846U

## TECHNICAL AND LABORATORY SKILLS

**Materials & Fabrication**: Poly(ionic liquid) elastomer synthesis, thiol-click chemistry, conductive-polymer processing, flexible electrode fabrication, screen printing, multilayer device integration.

**Characterization**: Universal mechanical testing, rheology, EIS, SEM, LCR measurements, sensor calibration, response/recovery analysis, limit-of-detection and cyclic-durability testing.

**Data Analysis & Visualization:** Origin, Python, Excel, FFT, Adobe Illustrator, Adobe Premiere Pro.

**Electronics & Embedded Systems**: Familiar with Altium Designer and STM32CubeIDE; collaborative experience in PCB prototyping and embedded signal acquisition.

## SELECTED AWARDS AND HONORS

### Academic & Research Excellence

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| Academic Scholarship, Southern University of Science and Technology | *2024-2025* |
| Academic Excellence Award, Chang'an University | *2021-2022; 2022-2023* |
| China Merchants Bank Scholarship, Chang'an University | *2020-2021* |

### Competitions and Leadership

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| Second Prize, National English Competition for College Students | *2021-2022* |
| Second Prize, National Mathematics Competition for College Students | *2022-2023* |
| Second Prize, Shaanxi Provincial Metallographic Skills Competition | *2022-2023* |

## LANGUAGES

Chinese (Native); English (TOEFL iBT: 108); German (Basic)